

CADSync AI Engineering Report

Part ID: AF_PANEL_004 | Revision: B

Part Summary

Key	Value
part_id	AF_PANEL_004
length	188.8
width	118.4
height	10.0
hole_count	10
hole_diameter	11.2
material	ABS_Polymer
tolerance	0.03
revision	B
hole_pattern	[{'Hole_ID': 'H1', 'X': 20.0, 'Y': 20}, {'Hole_ID': 'H2', 'X': 85.333, 'Y': 20}, {'Hole_ID': 'H3', 'X': 150.666, 'Y': 20}, {'Hole_ID': 'H4', 'X': 216.0, 'Y': 20}, {'Hole_ID': 'H5', 'X': 281.333, 'Y': 20}, {'Hole_ID': 'H6', 'X': 346.666, 'Y': 20}, {'Hole_ID': 'H7', 'X': 412.0, 'Y': 20}, {'Hole_ID': 'H8', 'X': 477.333, 'Y': 20}, {'Hole_ID': 'H9', 'X': 542.666, 'Y': 20}, {'Hole_ID': 'H10', 'X': 608.0, 'Y': 20}]
features	[{'Feature_Type': 'Fillet', 'Value': 1.5}, {'Feature_Type': 'Extrusion', 'Value': 10}, {'Feature_Type': 'Hole', 'Value': 11.2}, {'Feature_Type': 'Slot', 'Value': 2.0}, {'Feature_Type': 'Thread', 'Value': 0.5}, {'Feature_Type': 'Chamfer', 'Value': 0.5}, {'Feature_Type': 'Bore', 'Value': 10.0}, {'Feature_Type': 'Flange', 'Value': 5.0}, {'Feature_Type': 'Rim', 'Value': 5.0}, {'Feature_Type': 'Base', 'Value': 5.0}]
configurations	[{'Config': 'Default', 'Export_STEP': True, 'Export_IGES': True, 'Export_DXF': True}, {'Config': 'Custom', 'Export_STEP': False, 'Export_IGES': False, 'Export_DXF': False}]
metadata	{'Designer': 'CADSync AI', 'Project': 'Air Filter Parts Test Suite', 'Structure_Type': 'Panel', 'Version': '1.0', 'Date': '2023-10-27'}

Validation Results

Key	Value
Valid	True
Errors	None
Warnings	None

AI Assessment

Key	Value
Anomaly Score	0.0342
Anomaly Flag	False
Manufacturability Risk	0
Risk Probability	2.80%
Estimated Manufacturing Cost	USD 33.66

Optimization Suggestions

Key	Value
Length	188.8
Width	118.4
Hole_Spacing	20.16

- Hybrid optimization executed: GA seed + SciPy refinement.
- Objective prioritizes reduced material usage and manufacturable spacing.

Generated Artifacts

Key	Value
STEP	C:\Users\Tharun B S\Documents\VS Code\CADSync-AI\outputs\AF_PANEL_004\
IGES	C:\Users\Tharun B S\Documents\VS Code\CADSync-AI\outputs\AF_PANEL_004\
STL	C:\Users\Tharun B S\Documents\VS Code\CADSync-AI\outputs\AF_PANEL_004\
DXF	C:\Users\Tharun B S\Documents\VS Code\CADSync-AI\outputs\AF_PANEL_004\
Model	C:\Users\Tharun B S\Documents\VS Code\CADSync-AI\outputs\AF_PANEL_004\

Engineering Knowledge Graph

CADSync AI Knowledge Graph - AF_PANEL_004

